

Jon Elezgarai Miguel

+34 688 855 772 | elezgaraiion@gmail.com | linkedin.com/in/jon-elezgarai-miguel | github.com/elezgaraiion | jon-elezgarai.vercel.app

EDUCATION

University of the Basque Country (UPV/EHU)

Donostia-San Sebastián, Spain

BSc in Artificial Intelligence

2022 – 2026

- Four-year degree covering Machine Learning, Deep Learning, NLP, Computer Vision, Big Data, Statistics, and Software Engineering
- Completed final-year thesis on **spatial reasoning in multimodal models**: comparing image-only, text-only and multimodal systems to identify where each modality adds value

Sapienza Università di Roma

Rome, Italy

Erasmus Exchange — ACSAI (BSc) & MSc in Artificial Intelligence

2025 – 2026 · First semester

- Coursework included Planning & Reasoning, Visual Analytics, Web & Software Architecture, and Cybersecurity

PROJECTS

Spatial Reasoning in Multimodal Models (Final-Year Thesis) | [View PDF](#)

2025 – 2026

- Extended the DecompSR benchmark with multimodal image representations and evaluated text-only, image-only, and vision-language models for spatial reasoning
- Designed and evaluated controlled experiments across multiple model architectures, analysing when visual information improves spatial reasoning performance
- Built the evaluation pipeline in Python using Hugging Face Transformers and PyTorch, including automated inference, post-processing and metric computation
- Evaluated fusion strategies to identify where combining modalities genuinely outperforms unimodal approaches

Multilingual Document Translation and Classification | [View PDF](#)

2025

- Evaluated BERT-based and multilingual language models for Basque document classification using machine translation and cross-lingual transfer strategies
- Explored zero-shot and few-shot prompting approaches and evaluated their impact on classification performance

Parkinson's Disease Detection | [View PDF](#)

2025

- Developed and evaluated machine learning models for Parkinson's disease detection using clinical data
- Compared classification approaches and analysed model performance across different experimental configurations

Age Estimation from Images | [View PDF](#)

2025

- Developed a computer vision pipeline for age estimation from facial images using deep learning techniques
- Performed image preprocessing, model training and evaluation to assess age prediction performance

WASAText — Full-Stack Messaging Application | [GitHub](#)

2026

- Developed a full-stack web messaging application inspired by WhatsApp Web using **Go**, **Vue.js 3**, and **SQLite**
- Implemented private and group conversations with **text and photo messaging, replies, forwarding, emoji reactions, message deletion, and pagination**
- Built a modular client-server architecture around an **OpenAPI 3.0 REST API**, with Bearer Token authentication, state management, and containerized frontend/backend services

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java

Machine Learning / AI: PyTorch, scikit-learn, Transformers, pandas, NumPy, Weights & Biases

Big Data & Data Engineering: Apache Spark, Apache Kafka, Hadoop, MongoDB

Developer Tools: Git, Docker, Linux, Jupyter, VS Code

LANGUAGES

Spanish: Native

Basque: Native

English: B2 — Cambridge First Certificate (FCE)